

UNCERTAINTY-AWARE EXPLAINABLE MEDICAL IMAGE SEGMENTATION USING CASF-NET

Proposal ISP

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1.INTRODUCTION

1.1Background

Artificial intelligence has altered a number of industries, including the health care sector. AI is used in a few different ways within the health care realm but one of the biggest applications of AI in this industry is the analysis of medical images. The ability to use deep learning or CNN to identify disease through analysis of MRI/CT/X-ray/ultrasound images has helped to significantly advance the accuracy and efficiency of diagnosing medical conditions.

Medical image segmentation is the process of identifying and separating regions of interest (e.g tumors or organs etc.) from the rest of the medical image. Typically, CNNs have been the most widely used model for this process due to their ability to automatically learn features of an image. However, CNNs focus primarily on local image data and do not have a strong ability to capture long range context.

In order to resolve this shortcoming, researchers have introduced transformer architectures which provide models with the ability to learn global relationships or long-range dependencies within the data by implementing attention mechanisms.

Recently , there have been hybrid architectures for example, CASF-Net that combine the local features of CNNs with the global features of transformers. Crossing both CNN and transformer modalities during segmentation will provide an additional level of fusion in order to improve the performance of segmentation. As a result of these innovations, hybrid architectures are likely to be vehicle through which medical professionals will be able to continue developing accurate and reliable predictions.

While hybrid architectures are providing significant improvements in medical imaging segmentation, currently they still function as “black boxes” because it is difficult for medical professionals to ascertain how and why not the prediction was made and what the models level of confidence is with the output.

It is critical that all healthcare applications not just have acceptable levels of trust, but also be trustworthy in terms of both explainability and reliability. To this end, this study proposes using a combination of Monte Carlo Dropout (for estimating prediction uncertainty), along with Grad-CAM (as a method of generating visual explanations) integrated with CASF-Net to produce accurate image segmentations with associated confidence levels along with visual explanations, thus providing a much higher level of trust in AI-assisted diagnoses in the medical field.

1.2 Aim

This research intends to create an uncertainty-aware/explainable medical image segmentation framework based on CASF-Net to ultimately improve AI's reliability, transparency, and trustworthiness when being used to assist with the diagnosis of patients.

1.3 Objectives

The specific objectives of this research include:

- 1.) Investigating existing medical image segmentation techniques.
- 2.) Analyzing the architecture and performance of CASF-Net.
- 3.) Implementing CASF-Net for medical image segmentation.
- 4.) Integrating monte carlo dropout into CASF-Net for estimating prediction uncertainty.
- 5.) Utilizing Grad-CAM within CASF-Net for visual explanation of segmentations produced.
- 6.) Evaluating the performance of the proposed framework using standard segmentation performance metrics.
- 7.) Comparing the performance of the proposed framework against traditional segmentation techniques.

1.4 Description of final product

The final product will be a deep-learning based medical image segmentation system. This system will take a medical image as input and produce one or more segmented outputs. In addition to generating segmented outputs, the system will utilize Monte Carlo Dropout to estimate the predictive confidence of all segmentations produced via CASF-Net and provide visual explanations of the segmentations produced by CASF-Net using Grad-CAM.

Framework consist of:

- CASF-Net for segmentation
- Monte carlo dropout for uncertainty estimation
- Grad-CAM for visual explainability

So this final output will be included with segmentation masks, uncertainty scores, and heatmaps indicating the regions that influenced the models decisions.

1.5 Scope

This research covers:

- Medical image segmentation
- Explainable AI methods
- Techniques for estimating uncertainty
- Evaluating methods with publicly available data sets

Excluded area:

- Clinical use in real time

- Integration with hospital systems
- Collection of data from patients
- Clinical validation of health care providers

1.6 Targeted Audience

This framework will assist in:

- Researchers in medical imaging
- Researchers in artificial intelligence
- Health care providers
- Students studying artificial intelligence and medical imaging
- Developers of trustworthy AI systems

2. BACKGROUND REVIEW

2.1 Existing Techniques

The approaches to medical segmentation has progressed greatly over time.

Convolutional Neural Network Methods:

Models such as U-Net and ResUNet are built upon learning local image features through convolutions.

Advantages:

- Excellent at extracting local features from images
- Able to find boundaries effectively

Disadvantages:

- Limited ability to model long-range dependencies

Transformer methods:

Model such as vision transformer (ViT) and TransUNet use global attention mechanisms for learning about the context in the whole image.

Advantages:

- Able to learn about the whole global image

Disadvantages:

- Computationally intensive to work with
- May lose fine detail in the images

Hybrid Methods:

Models such as CASF-Net, TSCA-Net and MACCoM use both CNNs and Transformers.

Advantages:

- Able to learn local and global information
- Improved segmentation results

Disadvantages:

- Limited explainability of the results
- No ability to estimate uncertainty in the results

2.2 Literature Review

CASF-Net :

CASF-Net represents a dual-branch design consisting of a CNN and transformer architectures. Cross-attention and Cross-scale fusion unite local and global information sources effectively.

Advantages:

- Commonly exhibits high segmentation accuracy

Disadvantages:

- Does not provide uncertainty estimation or model explainability

TSCA-Net :

TSCA-Net captures channel and spatial attention features to enhance segmentation performance.

Advantages:

- Captures salient and important regions and features.

Disadvantages:

- Provides no uncertainty estimation

Grad-CAM:

It provides visual explanations by emphasizing the contributions of different image regions to the model's predictions, resulting in intuitive visual representations.

Advantages:

- A clear visual representation of image explanations.

Monte carlo Dropout:

It can provide an estimate of model uncertainty through use of multiple stochastic forward pass callbacks.

Advantages:

- Provides predictive confidence

3.METHODOLOGY

3.1 Research Design

The research proposes to follow a Design and Development research strategy methodology.

Step 1: Conduct literature review

Step 2: Select datasets and preprocess them

Step 3: Implement CASF-Net

Step 4: Add Monte Carlo Dropout and implement it.

Step 5: Add Grad-CAM and implement it.

Step 6: Evaluate the model using metrics such as simple accuracy, segmentation accuracy, true positive rate, false positive rate, mean absolute error (MAE), root mean square error (RMSE), and correlation of predicted values and ground truth.

Step 7: Analyse , report the results of the study.

3.2 Technology

Programming language:

- Python

Frameworks:

- PyTorch
- TensorFlow

Libraries:

- OpenCV
- NumPy
- Matplotlib

Environment:

- Google Colab
- Jupyter Notebook

Version Control:

- GitHub

3.3 Version Management Plan

GitHub will be used to manage:

- Source code
- Model versions
- Experiment results
- Documentation

Version milestones:

V1.0-CASF-Net implementation

V2.0-CASF-Net + MC Dropout

V3.0_CASF-Net + MC Dropout + Grad-CAM

4. PROJECT PLAN

4.1 Activities

- Literature review
- Dataset preparation
- Model implementation
- Uncertainty integration
- Explainability integration
- Evaluation
- Dissertation writing

4.2 Gantt chart

Activities	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
Topic selection and initial research	*								
Literature review	*	*	*						
Research Gap Identification		*	*						
Proposal preparation			*	*	*				
Ethics Form Preparation				*	*				

Dataset collection						*			
Data preprocessing						*	*		
CASF-Net implementation						*	*		
Monte Carlo Dropout integration							*		
Grad-CAM integration							*	*	
Model Training and Testing							*	*	
Performance Evaluation								*	
Dissertation writing							*	*	*
Presentation preparation								*	*
Dissertation presentation									*
Final corrections and submission									*

4.3 Data Management plan

All project files will be stored securely with help of:

- Google drive
- GitHub
- Local backups

Files include:

- Research papers
- Datasets
- Source code
- Results
- Dissertation drafts.

4.4 Deliverables

1. Literature review

2. Implemented CASF-Net model
3. Monte Carlo Dropout module
4. Grad-CAM module
5. Experimental results
6. Dissertation report
7. Presentation slides
8. GitHub repository

PROFESSIONAL ISSUES

Ethical Issues

We use only publicly available datasets and no other human participants will be involved.

Legal Issues

Every datasets and research papers will be cited and used according to the licensing agreements.

Social Issues

Every AI systems should be transparent and trustworthy to encourage adoption in healthcare.

Environmental Issues

Computational resources and energy was consumed by Deep learning training

Sustainability Issues

The framework will be designed to be reusable and reproducible for future research

Risk Management

The potential risks are dataset limitations, computing power issues, and overfitting of the models being used. These risks have appropriate mitigation strategies including data augmentation, Dropout, and cloud based computing resources.